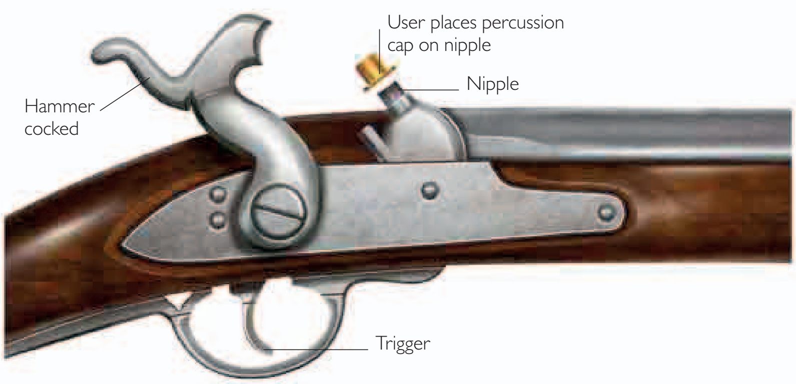
HOW GUNS WORK

FROM THE 19TH CENTURY

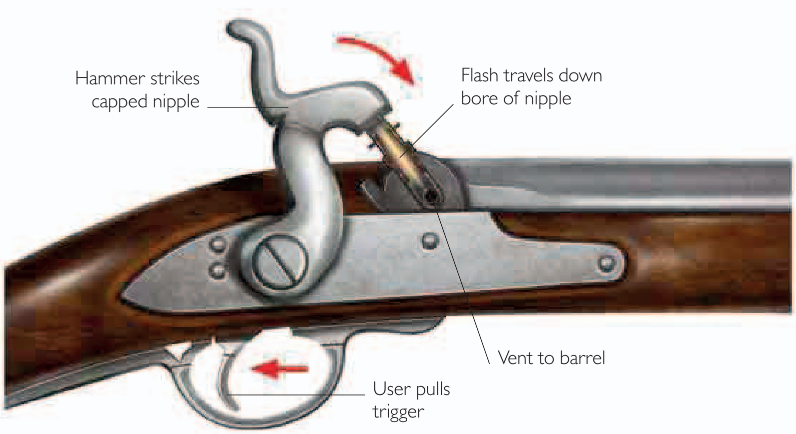
The invention of percussion caps provided firearms with an instantaneous method for the chemical ignition of the propellant (gunpowder). By the 1870s, these caps were contained within fully integrated metallic cartridges. These cartridges carried a projectile, propellant, and a primer in one compact package. Cartridges could be loaded quickly at the breech of the gun—with the cartridges being fed into the chamber by bolt action. Soon, cartridges were being fed repeatedly from magazines. The automation of this loading process, from magazines or belts, using a recoil-operated or a gas-operated action, led to semi-automatic (self-loading) and fully automatic weapons.

Percussion cap

A percussion cap is formed of two layers of copper foil with a mixture of fulminate of mercury, potassium chlorate, and sulfur or antimony between them. The composition catches fire when the hammer strikes it.



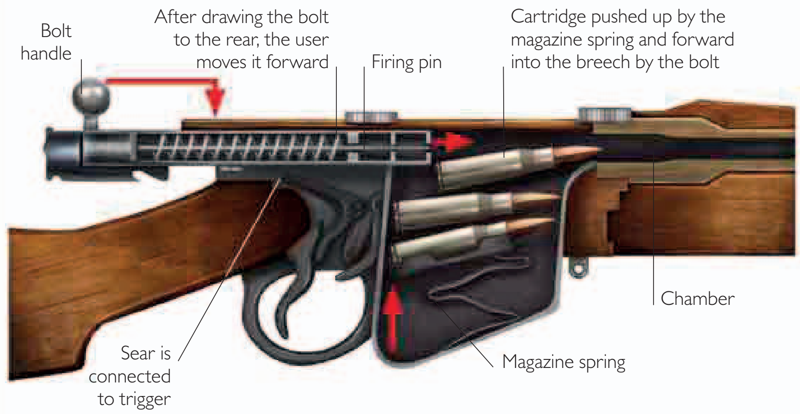
1 A sear (a hooklike part inside the gun) holds the hammer in the cocked position. The sear connects to the trigger. The user places the percussion cap on the nipple, the bore of which leads to the propellant in the barrel.



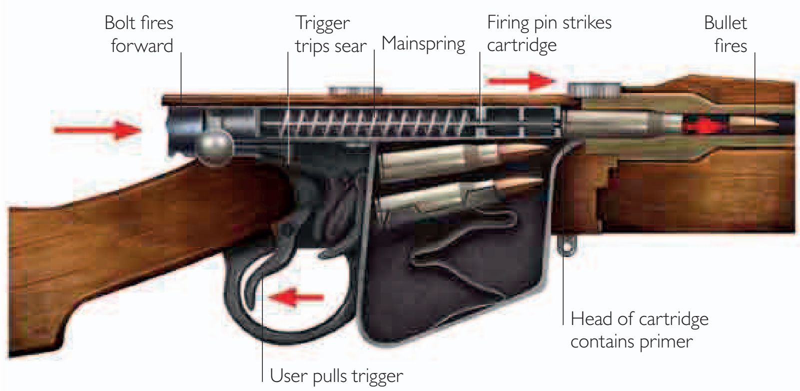
2 Pulling the trigger trips the sear, releasing the hammer and driving it onto the nipple. The primer in the cap ignites. The flame passes down the bore in the nipple and through a vent into the main charge in the barrel, igniting it.

Bolt action

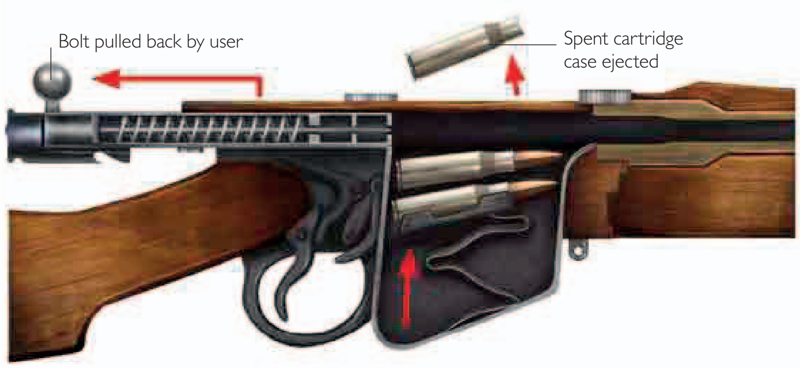
Bolt action, essentially based on the device that holds a garden gate closed, is a sure and effective design of breech-loading firearm. The mechanism was used with the first repeater rifles, which were the first guns with magazines. The magazines contained cartridges ready to be loaded and fired.



1 The user lifts the bolt handle, rotating the body of the bolt and freeing its locking lugs, and draws it fully to the rear. This opens the breech of the gun. As the user moves the bolt forward, it picks up a cartridge from the magazine and chambers it.



2 As the user returns the bolt handle to the closed position, seating the locking lugs and sealing the breech, the mainspring and firing pin are held back by the sear, which keeps the bolt cocked. Pulling the trigger trips the sear and releases the firing pin. As the mainspring decompresses, the pin flies forward and impacts the primer at the head of the cartridge, detonating it and firing the bullet.



3 As the user withdraws the bolt, it extracts the spent cartridge case by means of a hook on the bolt head, which engages with the rim of the case. The recoil force generated pushes the bolt backward, compressing the mainspring, which then springs forward once more. The movement pushes up the next cartridge.